

Natural_Resources_Price_Trend_Analysis

Introduction

Natural resources play a central role in global economic activity, shaping everything from industrial production to long-term investment decisions. Their prices shift over time in response to changes in supply, demand, geopolitical tensions, and broader economic conditions. This project looks at how the prices of key natural resources have evolved over the years and uses forecasting methods to explore what those trends might mean for the future.

Business Problem

Commodity prices move unpredictably, and these fluctuations can have major consequences for governments, companies, and investors. Being able to recognize long-term patterns, seasonal behavior, and structural shifts is essential for making informed decisions. This project focuses on understanding how commodity prices have evolved over time and what those historical trends can tell us about future market movements.

Objectives

This project focuses on understanding long-term movements in natural resource prices and building a practical forecasting workflow. The key objectives are to:

- **Analyze long-term price data** — review natural resource prices from 2000 to 2023.
- **Identify seasonality patterns** — explore weekly, monthly, and yearly cycles.
- **Visualize price behavior** — create charts showing daily, weekly, and monthly movements.
- **Examine trend horizons** — use 7-, 30-, and 365-day rolling averages to capture short-, medium-, and long-term trends.
- **Build a predictive model** — develop and evaluate a forecasting model using cleaned and prepared data.

- **Produce a multi-year forecast** — generate a three-year outlook for selected natural resources.
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Dataset Description

The dataset brings together more than two decades of daily price data for a wide range of natural resources. It spans energy commodities, metals, agricultural products, and livestock, with each entry capturing the date and the market price for each resource. This long-term dataset makes it possible to study patterns, trends, and shifts across different commodity groups.

Methodology

All data preparation and analysis were done in Python within Jupyter Notebook. After cleaning the dataset, I resampled the data to multiple time frequencies to study seasonality and broader price behaviour, using line and box plots to visualise the patterns. I also explored the structure of the data through autocorrelation and experimented with models such as decision trees and random forests. Rolling averages were useful for separating short-term fluctuations from longer-term trends. To forecast future prices, I used Prophet to build models that produced three-year projections for key commodities.

Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Plotly
- Prophet
- Jupyter Notebook

Key Findings

The analysis revealed several important insights into how natural resource prices behave over time:

- **Price volatility** — Natural resource prices showed significant fluctuations throughout the study period.
 - **Distinct commodity behaviors** — Each commodity group displayed its own long-term trends and seasonal patterns.
 - **Rolling-average clarity** — Rolling averages helped highlight underlying movements that were not immediately visible in the raw data.
 - **Forecasting outcomes** — Predictive models indicated continued price changes ahead, with uncertainty increasing the further the forecast extended.
 - **Role of historical patterns** — Long-term trends and seasonality played a meaningful role in improving forecast accuracy.
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Limitations

While the analysis provides meaningful insights, several limitations should be noted:

- **Historical-data dependence** — The forecasts rely heavily on past price behaviour and do not explicitly account for geopolitical shocks, economic crises, or supply-chain disruptions.
 - **Declining accuracy over time** — Forecast reliability decreases as the projection period extends.
 - **External shocks** — Unexpected events or structural market changes may cause actual prices to diverge from model predictions.
 - **Some complexity is unavoidable** — Commodity markets are influenced by many variables that are difficult to capture fully in a single model.
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Conclusion

This project shows how Python and time-series techniques can be used to explore long-term commodity price behavior, uncover seasonal patterns, and build practical forecasting models. By combining visual analysis, trend exploration, and predictive modelling, the study offers meaningful insights into how natural resource prices evolve over time. The results highlight the value of data analytics in supporting economic planning, market analysis, and investment decision-making.